Notes: Acemoglu and Restrepo (JPE, 2020)
Robots and Jobs: Evidence from US Labor Markets

Contents

1 Big picture

- **Question.** What is the equilibrium effect of industrial robots on local US employment and wages? The paper asks whether robot adoption is just another form of productivity-enhancing capital deepening, or whether it has a distinctive labor-displacing effect.
- **Core idea.** Robots replace workers in specific tasks. This creates a **displacement effect** that can reduce employment and wages. At the same time, lower production costs create a **productivity effect** that can raise demand. The empirical question is which force dominates in US labor markets.
- **Setting.** The paper studies US commuting zones from 1990 to 2007, a period in which industrial robot use rose rapidly in the United States and Europe. The analysis links local labor market outcomes to exposure to industries where robots became technologically feasible and widely adopted.
- **Main measure.** The key regressor is exposure to robots, a shift-share measure combining a commuting zone's baseline industry employment shares with industry-level robot penetration. The preferred shock uses robot adoption in advanced European economies to isolate robot technology trends rather than US-specific local shocks.
- **Headline result.** Commuting zones more exposed to robots experience lower employment-to-population ratios and lower wages. The paper reports that one more robot per thousand workers reduces the aggregate employment-to-population ratio by about 0.2 percentage points and wages by about 0.42%.
- **Local versus aggregate effects.** Local effects are larger than aggregate effects because robots in one commuting zone also lower prices of tradable goods and generate spillovers to other areas. The paper therefore distinguishes local reduced-form estimates from aggregate implications.
- **Heterogeneity.** Negative employment effects are concentrated in manufacturing, routine manual, blue-collar, assembly, and related occupations. The wage effects are also visible across the wage distribution and by education and gender.
- **Economic takeaway.** Robots are not equivalent to ordinary capital accumulation. They are a specific automation technology that can reduce labor demand through task displacement, even when they raise productivity.

2 Incremental contribution of the paper

- **From automation concern to equilibrium evidence.** Earlier discussions of automation often relied on task forecasts or occupation-level susceptibility measures. This paper provides direct evidence on realized robot adoption and equilibrium local labor market outcomes.
- **Theory-guided empirical design.** The paper builds a task-based model in which robots compete with labor in production. The model delivers the empirical exposure variable and clarifies why automation can differ from capital deepening.
- **Industrial robots as a measurable automation technology.** The International Federation of Robotics data allow the authors to measure actual robot stocks by industry and country, rather than using indirect proxies for automation.

- **Shift-share identification with European robot adoption.** The paper uses industry-level robot adoption in European countries that are ahead of the United States in robotics as the source of technological variation. This helps separate global robot technology trends from endogenous US industry shocks.
- **Local and aggregate interpretation.** The paper goes beyond estimating local reduced-form effects. It uses the model to translate local estimates into aggregate employment and wage effects, accounting for spillovers across commuting zones.
- **Robots versus other technologies.** A central contribution is to show that the negative employment and wage effects are not replicated by overall capital accumulation, IT capital, or industry value added growth.
- **Distributional contribution.** The paper documents which sectors, occupations, education groups, genders, and wage quantiles are most affected, giving the study a distributional and policy-relevant dimension.

3 Data and measurement

3.1 Robot data

- The paper uses data from the International Federation of Robotics (IFR), which reports the stock of industrial robots by industry, country, and year.
- An industrial robot is an automatically controlled, reprogrammable, multipurpose machine that can perform tasks such as welding, painting, assembly, handling, and packaging without a human operator.
- The paper focuses on 19 IFR industries: 13 manufacturing industries and 6 broad nonmanufacturing industries.
- US robot data by industry are limited before 2004, so the preferred exposure measure is based on robot penetration in a set of advanced European economies.
- The main European measure uses Denmark, Finland, France, Italy, and Sweden, excluding Germany in the baseline because Germany is far ahead of the other countries in robot density.

3.2 Commuting zone data

- The local labor market unit is the commuting zone. The sample covers 722 commuting zones in the continental United States.
- Employment, population, wages, demographic characteristics, industry shares, and occupation measures come from the Census, ACS, CBP, BEA, IRS, and related data sources.
- The baseline local industry structure is measured before the main robot adoption period. In the preferred exposure measure, baseline employment shares come from 1970 to reduce mechanical links with later robot adoption.
- Wage outcomes are measured using demographic-cell by commuting-zone data, where cells are defined by gender, education, age, and race. This helps control for changes in observable worker composition.

3.3 Exposure to robots

The paper's central empirical measure is a Bartik-style exposure variable:

$$\text{Exposure to robots}_{c,(t_0,t_1)} = \sum_i \ell_{ci}^{1970} \cdot APR_{i,(t_0,t_1)},$$

where ℓ_{ci}^{1970} is the baseline share of commuting zone c's employment in industry i, and APR_i is the adjusted penetration of robots in industry i computed from European robot adoption.

The US analogue is

$$\text{US exposure to robots}_{c,(t_0,t_1)} = \sum_i \ell_{ci}^{1990} \cdot APR_{i,(t_0,t_1)}^{US}.$$

The US measure is used in IV specifications, while the European measure is used as the source of predicted robot exposure.

3.4 Other controls and comparison shocks

- The baseline controls include census division fixed effects, demographics, manufacturing and light-manufacturing employment shares, female manufacturing share, exposure to Chinese imports, and routine job share.
- Additional robustness specifications control for past local trends, local commuting-zone trends, exposure to capital deepening, exposure to IT capital, exposure to industry value added, and separate automotive versus nonautomotive robot exposure.
- These controls are designed to address the concern that robot-exposed areas might have been declining for unrelated reasons.

4 Models and empirical specifications

4.1 Task model of robots and labor

- Each industry produces using a continuum of tasks. Some tasks can be performed by robots; other tasks must be performed by labor.
- An increase in the set of robot-performable tasks generates a displacement effect because robots directly replace workers in those tasks.
- Lower production costs also generate productivity and composition effects, which may raise output and labor demand in nonautomated tasks.
- The total effect on employment and wages depends on whether the displacement effect dominates the productivity effect.

4.2 Reduced-form estimating equations

The main local reduced-form equations are:

$$\Delta \ln L_c = \beta_L \,\text{Exposure to robots}_c + X_c'\Gamma_L + \varepsilon_c^L,$$

$$\Delta \ln W_c = \beta_W \text{ Exposure to robots}_c + X_c'\Gamma_W + \varepsilon_c^W.$$

In the empirical tables, the employment outcome is usually the change in the employment-to-population ratio, while the wage outcome is the change in log hourly wages.

- A negative β_L means robot exposure reduces employment relative to population.
- A negative β_W means robot exposure reduces wages.
- The estimates are reported both in long differences and in stacked differences.

4.3 Long differences

The main long-difference specification compares local labor market changes from 1990 to 2007:

$$\Delta y_{c,1990-2007} = \beta \text{ Exposure}_{c,1993-2007} + X_c'\Gamma + \varepsilon_c.$$

This specification captures medium-run changes during the period when industrial robot adoption accelerated.

4.4 Stacked differences

The stacked-differences specification uses two periods, 1990-2000 and 2000-2007:

$$\Delta y_{ct} = \beta \text{ Exposure}_{ct} + X_c'\Gamma + \lambda_t + \varepsilon_{ct}.$$

This specification uses timing variation and can include commuting-zone trends or fixed effects.

4.5 Instrumental variables

The paper also estimates IV models in which US robot exposure is instrumented using European robot exposure:

$$\text{US exposure}_c = \pi \text{ European exposure}_c + X_c'\Pi + u_c.$$

This IV strategy addresses measurement error in US robot data and possible endogeneity of US robot adoption to US industry or local shocks.

5 Identification and empirical design

5.1 Identifying assumption

The core assumption is that commuting zones specializing in industries with greater robot technology advances would not have experienced different employment and wage trends for other reasons, conditional on the included controls.

5.2 Why European robot adoption helps

- Robot adoption in Europe proxies for industry-level advances in robotics technology.
- Using European adoption filters out US-specific shocks, such as local union pressure, local product demand shocks, or US industry-specific business-cycle forces.
- The first-stage idea is that industries adopting robots in Europe are also industries in which robot technology became feasible and attractive in the United States.

5.3 Threats to identification

- **Preexisting decline.** Robot-exposed commuting zones might already have been on a downward trend before robots spread.
- **Trade shocks.** Robot-exposed industries might overlap with industries hit by Chinese import competition or offshoring.
- **Automotive industry dominance.** The automotive industry is highly robot-intensive and geographically concentrated.
- **Other technologies.** Robot exposure might proxy for capital deepening, IT adoption, or productivity growth.
- **Shift-share concerns.** As with all Bartik designs, the source of variation is a combination of industry shocks and local industry composition.

5.4 How the paper addresses these concerns

- The paper shows no strong pretrends in employment and wages between 1970 and 1990 for later robot-exposed areas.
- The results are robust to controlling for Chinese import exposure, routine job share, industry trends, and past local outcome changes.
- The results survive excluding the top 1% of most-exposed commuting zones and separating the automotive industry from other industries.
- Capital deepening, IT capital, and value added do not explain away the robot effects.
- Industry-level evidence shows robot-adopting industries have lower labor demand and labor share but higher value added, matching the displacement-plus-productivity logic.

6 Tables and figures

6.1 Figure 1 and Figure 2: Robot adoption over time and across industries

Read:

- Figure 1 shows that robot density rose substantially in the United States and Europe after the early 1990s.

- Germany is far ahead of the United States and other European countries in robot density, which is why Germany is excluded from the baseline European shock measure.
- Figure 2 shows that industries with larger robot penetration in EURO5 also tend to have larger robot penetration in the United States.
- The most robot-intensive industries include automotive, electronics, plastics and chemicals, basic metals, and metal products.

Economic message: Robot adoption is a real, measurable, and industry-specific technological shock. The cross-industry correlation between European and US robot adoption motivates using European robot penetration as a source of predicted US exposure.

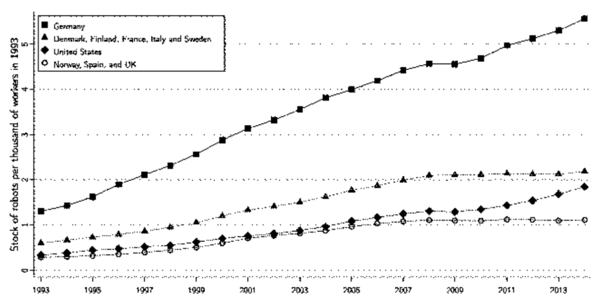


FIG. 1.—Industrial robots per thousand workers in the United States and Europe.

(a) Figure 1: Industrial robots per thousand workers in the United States and Europe.

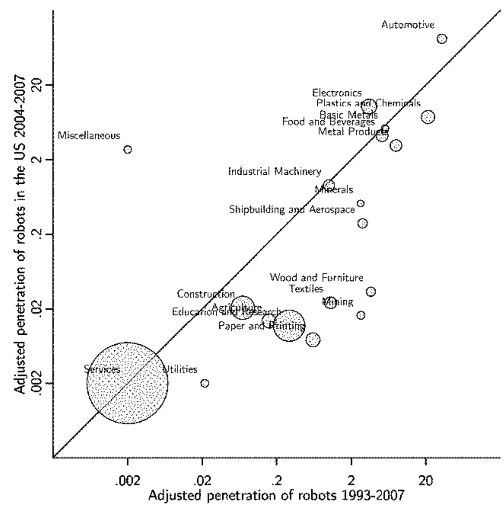


FIG. 2.—Adjusted penetration of robots in the United States and *EURO5* by industry. Plot of the adjusted penetration of robots between 1993 and 2007 ($\overline{APR}_i$) and the adjusted penetration of robots in the United States between 2004 and 2007 (APR_i rescaled to a 14-year equivalent change). Adjusted penetration of robots is given in number of robots per thousand workers in the industry. The solid line corresponds to the 45° line. Circle size indicates

(b) Figure 2: Adjusted penetration of robots in the United States and EURO5 by industry.

6.2 Table 1: Industry-level labor demand, labor share, and value added

Read:

- Table 1 relates industry-level robot penetration to changes in wage bill, employment, labor share, and value added.
- Robot adoption is associated with lower wage bills and lower employment, both in long differences and stacked differences.
- At the same time, robot adoption is associated with higher value added and a lower labor share.
- These patterns are consistent with robots raising productivity while making production less labor intensive.

Economic message: The industry-level evidence supports the theoretical mechanism: robots increase production capacity but displace labor from affected tasks, causing the labor share and employment to fall in robot-adopting industries.

TABLE 1

Robots, Labor Demand, Labor Share, and Value Added: Industry-Level Results

	Long Differences, 1993–2007				Stacked Differences, 1993–2000 and 2000–2007					Long Differences, 1992–2007
	CBP (All Industries)		NBER-CES (within Manufacturing)		CBP (All Industries)			NBER-CES (within Manufacturing)		BEA-IO
	All Workers (1)	All Workers (2)	All Workers (3)	Production Workers (4)	All Workers (5)	All Workers (6)	All Workers (7)	All Workers (8)	Production Workers (9)	All Workers (10)
	A. Change in Log Wage Bill									Value Added
Adjusted penetration of robots, $\overline{APR}_i$	−2.718	−.923	−.816	−.993	−2.510	−1.096	−1.492	−1.037	−1.150	.128
	(.732)	(.419)	(.378)	(.324)	(.673)	(.235)	(.481)	(.177)	(.205)	(.061)
Observations	19	19	13	13	38	38	38	26	26	19
R^2	.19	.91	.84	.91	.53	.90	.95	.87	.91	.72
	B. Change in Log Employment									Labor Share
Adjusted penetration of robots, $\overline{APR}_i$	−1.967	−.754	−.831	−.991	−1.904	−.883	−1.325	−.921	−1.016	−.797
	(.654)	(.347)	(.339)	(.261)	(.609)	(.129)	(.329)	(.114)	(.152)	(.281)

(a) Table 1, part 1: Industry-level estimates.

Observations	19	19	13	13	38	38	38	26	26	19
R^2	.12	.90	.87	.92	.30	.86	.93	.89	.93	.37
	Covariates									
Time period dummies					✓	✓	✓	✓	✓	
Broad industry dummies		✓	✓	✓		✓	✓	✓	✓	✓
Chinese imports		✓	✓	✓		✓	✓	✓	✓	✓
Industry fixed effects							✓			

Note.—This table presents estimates of the relationship between adjusted penetration of robots and the wage bill, employment, value added, and labor share across US industries. Columns 1–4 present long-differences estimates for changes in the log wage bill for 1993–2007 (panel A) and log employmen for 1993–2007 (panel B). Columns 5–9 present stacked-differences estimates for changes in the log wage bill for 1993–2000 and 2000–2007 (panel A) anc log employment for 1993–2000 and 2000–2007 (panel B). Column 10 presents long-differences estimates for changes in log value added for 1992–2007 (panel A) and labor share for 1992–2007 (panel B). Changes in log value added are annualized and given in percent change per year. Changes in labor share are in percentage points. Data sources and time periods are reported at the top of the table, and the set of covariates is reported at the bottom Column 1 does not include any covariates, and col. 5 includes only time period dummies. Columns 2–4, 6–9, and 10 control for dummies for manufac turing and light manufacturing (paper/printing and textiles) and exposure to Chinese imports by industry from Acemoglu et al. (2016). Column 7 in cludes a full set of industry fixed effects. The regressions in cols. 1–9 are weighted by baseline industry employment in 1993, and the regressions in col. 1(are weighted by baseline value added by industry in 1992. Standard errors that are robust against heteroskedasticity and serial correlation at the industry level are given in parentheses.

(b) Table 1, part 2: Covariates and notes.

6.3 Figure 3 and Figure 4: Industry labor demand and local exposure

Read:

- Figure 3 plots residualized industry-level relationships between robot penetration and changes in wage bill and employment.
- The fitted lines slope downward: industries with more robot penetration experience larger declines in wage bill and employment.
- Figure 4 maps commuting-zone exposure to robots. Exposure is highest in parts of the Midwest, South, Texas, and other manufacturing-intensive areas.
- Panel B shows that there remains substantial spatial variation even after excluding the automotive industry.

Economic message: Robots matter both at the industry level and spatially. Because industries are unevenly distributed across space, a sectoral robot shock becomes a geographically uneven local labor market shock.

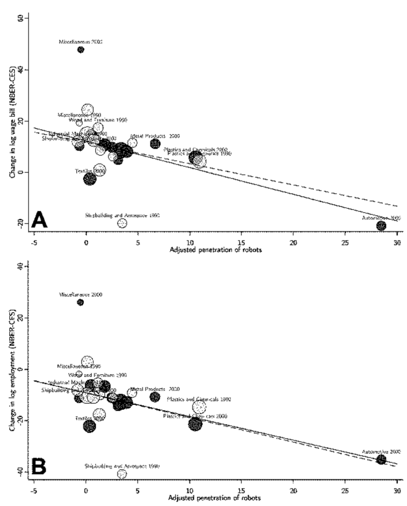
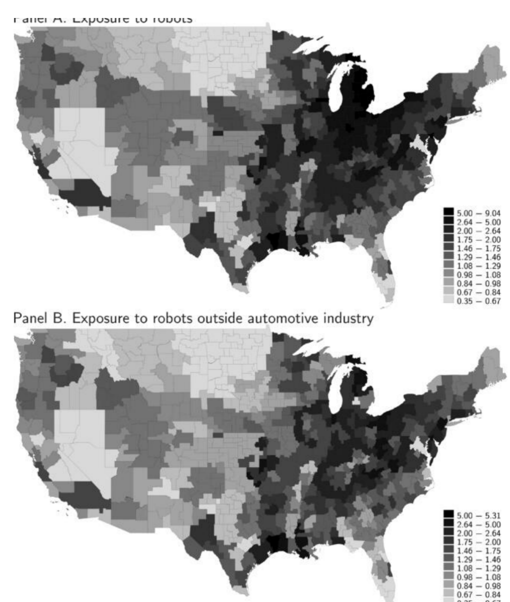


Fig. 3.—Relationship between robots and labor demand across industries. This figure presents residual plots of the relationship between adjusted penetration of robots ($\overline{APR}_i$) and the change in log wage bill (A) and the change in log employment (B) from stacked-differences models, with data for 1993–2000 (in light gray) and 2000–2007 (in dark gray). The solid line shows the coefficient estimates from column 8 of panel A (A) and column 8 of panel B (B) of table 1. The covariates from these models are partialed out. The dashed line is for a regression that additionally excludes the automotive industry. Circle size indicates the

(a) Figure 3: Robots and industry labor demand.



(b) Figure 4: Geographic distribution of exposure to robots.

6.4 Figure 5 and Table 2: Local adoption evidence and main long-difference results

Read:

- Figure 5 shows that robot-exposed commuting zones have more robot integrator activity, which is indirect evidence that the exposure measure is related to actual local robot activity.
- Table 2 reports the main long-difference reduced-form estimates for 1990-2007.
- In the preferred weighted specification with controls, one more robot per thousand workers is associated with about a 0.448 percentage point reduction in the employment-to-population ratio.
- The corresponding wage estimate is about a 0.884% reduction in hourly wages.
- The effects remain negative when excluding the most exposed commuting zones and in unweighted regressions.

Economic message: The central reduced-form finding is negative and robust: commuting zones more exposed to robots lose employment and wages relative to less exposed areas.

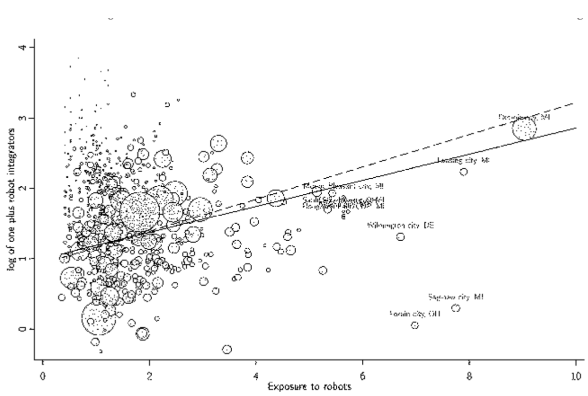


Fig. 5.—Exposure to robots and the location of robot integrators. This figure presents the relationship between exposure to robots for 1993–2007 and the log of one plus the number of robot integrators in a commuting zone. The covariates from column 4 of table 2 are partialed out. Data on the location of robot integrators are from Leigh and Kraft (2018). The solid line corresponds to a regression with the commuting zone population in 1990 as weights. The dashed line is for a regression that additionally excludes the top 1% of commuting zones with the highest exposure to robots. Circle size indicates the 1990 population in the commuting zone.

(a) Figure 5: Exposure to robots and robot integrators.

TABLE 2
Effects of Robots on Employment and Wages: Long Differences

	Long Differences, 1990–2007					
	Weighted by Population				Excludes Zones with the Highest Exposure	Unweighted
	(1)	(2)	(3)	(4)	(5)	(6)
	A. Change in Employment-to-Population Ratio, 1990–2007					
Exposure to robots	−.445	−.414	−.434	−.448	−.572	−.516
	(.094)	(.076)	(.057)	(.059)	(.138)	(.118)
Observations	722	722	722	722	712	722
R^2	.27	.46	.66	.67	.66	.62
	B. Change in Log Hourly Wages, 1990–2007					
Exposure to robots	−1.220	−1.017	−.874	−.884	−.779	−.932
	(.163)	(.126)	(.134)	(.132)	(.274)	(.205)
Observations	87,100	87,100	87,100	87,100	85,776	87,100
R^2	.32	.33	.33	.33	.33	.08
	Covariates					
Census divisions	✓	✓	✓	✓	✓	✓
Demographics		✓	✓	✓	✓	✓
Industry shares			✓	✓	✓	✓
Trade, routine jobs				✓	✓	✓

Note.—This table presents estimates of the effects of exposure to robots on employment and wages. Panel A presents long-differences estimates for changes in the employment-to-population ratio for 1990–2007. Panel B presents long-differences estimates for changes in log hourly wages for 1990–2007. The specifications in panel B are estimated at the demographic cell × commuting zone level, where demographic cells are defined by age, gender, education, and race. Columns 1–5 present regressions weighted by population in 1990. Column 5 presents results excluding the top 1% of commuting zones with the highest exposure to robots. Column 6 presents unweighted regressions. The covariates included in each model are reported at the bottom of the table. Column 1 includes only census division dummies. Column 2 adds demographic characteristics of commuting zones in 1990 (log population; the share of females; the share of the population over 65 years old; the shares of the population with no college, some college, college or professional degree, and masters or doctoral degree; and the shares of whites, blacks, Hispanics, and Asians). Column 3 adds the shares of employment in manufacturing and light manufacturing and the female share of manufacturing employment in 1990. Columns 4–6 add exposure to Chinese imports and the share of employment in routine jobs. Standard errors that are robust against heteroskedasticity and correlation within states are given in parentheses.

(b) Table 2: Effects of robots on employment and wages, long differences.

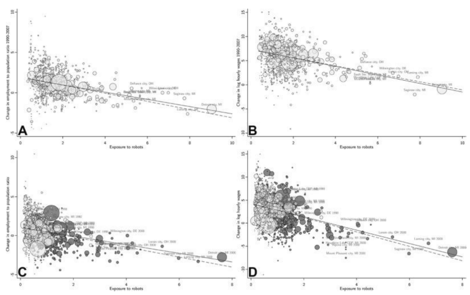
6.5 Figure 6 and Table 3: Visual reduced-form evidence and stacked differences

Read:

- Figure 6 plots residualized local relationships between robot exposure and changes in employment and wages in long differences.
- All panels show negative slopes, and the patterns persist when excluding the most exposed commuting zones.

- Table 3 uses stacked differences for 1990-2000 and 2000-2007.
- The stacked estimates are also negative. The preferred specifications imply sizable losses in employment-to-population ratios and wages.
- Including commuting-zone trends makes the wage effect even more negative in the unweighted specification.

Economic message: The results are not driven by a single long-difference comparison. Timing variation across two periods points in the same direction as the baseline design.



6.—Effects of robots on employment and wages. *A*, *B*, Long-differences relationship between exposure to robots and changes in the employment-ulation ratio for 1990–2007 (*A*) and log hourly wages for 1990–2007 (*B*). The covariates from column 4 of table 2 are partialed out. *C*, *D*, Stacked-nces relationships between exposure to robots and changes in the employment-to-population ratio (*C*) and log hourly wages (*D*) for 1990–2000 00–2007. In *C* and *D*, the observations for 1990–2000 are shown in light gray, the observations for 2000–2007 are shown in dark gray, and the co-s from column 4 of table 3 are partialed out. In all panels, the solid line shows the coefficient estimate from a regression with the commuting zone ıtion in 1990 as weights. The dashed line is for a regression that additionally excludes the top 1% of commuting zones with the highest exposure to . Circle size indicates the 1990 population in the commuting zone.

(a) Figure 6: Long-difference relationships between robot exposure and labor market outcomes.

TABLE 3
Effects of Robots on Employment and Wages: Stacked Differences

	Stacked Differences, 1990–2000 and 2000–2007							
	Weighted by Population				Excludes Zones with Highest Exposure	Unweighted	Commuting Zone Trends: Weighted by Population	Commuting Zone Trends: Unweighted
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	A. Change in Employment-to-Population Ratio, 1990–2000 and 2000–2007							
Exposure to robots	−.625	−.591	−.525	−.551	−.702	−.743	−.508	−1.007
	(.092)	(.076)	(.059)	(.052)	(.150)	(.092)	(.079)	(.116)
Observations	1,444	1,444	1,444	1,444	1,424	1,444	1,444	1,444
R^2	.24	.32	.39	.41	.40	.39	.71	.44
	B. Change in Log Hourly Wages, 1990–2000 and 2000–2007							
Exposure to robots	−1.544	−1.508	−1.405	−1.443	−1.643	−1.684	−1.608	−2.649
	(.211)	(.199)	(.191)	(.182)	(.551)	(.295)	(.271)	(.407)
Observations	183,606	183,606	183,606	183,606	180,818	183,606	183,606	183,606

(b) Table 3, part 1: Stacked-difference estimates.

6.6 Table 3 continued and Figure 7: Pretrends

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- The continuation of Table 3 reports covariates and notes for the stacked-difference estimates.
- Figure 7 directly checks whether areas later exposed to robots were already experiencing negative employment or wage trends from 1970 to 1990.
- The figure shows no strong negative pretrend in either employment-to-population ratios or hourly wages.
- This is important because the main estimates would be less credible if robot-exposed areas had been declining before robot adoption accelerated.

Economic message: The pretrend evidence supports the interpretation that the post-1990 results are tied to robot exposure rather than simply continuing earlier local decline.

	Covariates							
Time period dummies	✓	✓	✓	✓	✓	✓	✓	✓
Census divisions	✓	✓	✓	✓	✓	✓	✓	✓
Demographics		✓	✓	✓	✓	✓	✓	✓
Industry shares			✓	✓	✓	✓	✓	✓
Trade, routine jobs				✓	✓	✓	✓	✓
Commuting zone trends							✓	✓

NOTE.—This table presents estimates of the effects of exposure to robots on employment and wages. Panel A presents stacked-differences estimates fo changes in the employment-to-population ratio for 1990–2000 and 2000–2007. Panel B presents stacked-differences estimates for changes in log hourl wages for 1990–2000 and 2000–2007. The specifications in panel B are estimated at the demographic cell × commuting zone level, where demographi cells are defined by age, gender, education, and race. Columns 1–5 and 7 present regressions weighted by population in 1990. Column 5 presents result excluding the top 1% of commuting zones with the highest exposure to robots. Columns 6 and 8 present unweighted regressions. The covariates include in each model are reported at the bottom of the table. Column 1 includes only census division dummies and time period dummies. Column 2 adds de mographic characteristics of commuting zones (log population; the share of females; the share of the population over 65 years old; the shares of th population with no college, some college, college or professional degree, and masters or doctoral degree; and the shares of whites, blacks, Hispanic and Asians). Column 3 adds the shares of employment in manufacturing and light manufacturing and the female share of manufacturing employmen Columns 4–6 add exposure to Chinese imports and the share of employment in routine jobs. In addition, cols. 7 and 8 include a full set of commutin zone fixed effects. Standard errors that are robust against heteroskedasticity and correlation within states are given in parentheses.

(a) Table 3, part 2: Covariates and notes.

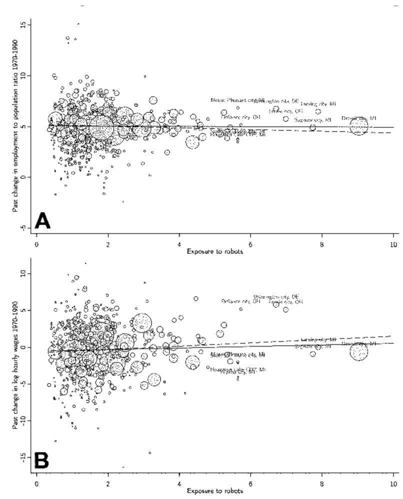


FIG. 7.—Pretrends in employment and wages. This figure presents the relationship between exposure to robots for 1993–2007 and the change in the employment-to-population ratio for 1970–1990 (*A*) and the change in log hourly wages for 1970–1990 (*B*). The covariates from column 2 of table 4 are partialed out. The solid line shows the coefficient estimate from a regression with the commuting zone population in 1990 as weights. The dashed line is for a regression that additionally excludes the top 1% of commuting zones with highest exposure to robots. Circle size indicates the 1970 population in the commuting zone.

(b) Figure 7: Pretrends in employment and wages.

6.7 Table 4: Pretrend controls

Read:

- Table 4 formalizes the pretrend tests and controls.
- Panel A uses past changes in labor market outcomes from 1970 to 1990 as dependent variables and finds no robust evidence that future robot exposure predicts prior declines.
- Panels B and C show that the main post-1990 robot effects remain after controlling for earlier industry trends or earlier local changes in the dependent variables.

Economic message: The estimates are not simply picking up long-running deterioration in robot-exposed areas. Controlling for earlier trends leaves the robot effects largely intact.

TABLE 4
EFFECTS OF ROBOTS ON EMPLOYMENT AND WAGES: TESTING AND CONTROLLING FOR PRETRENDS

	CHANGE IN EMPLOYMENT-TO-POPULATION RATIO				CHANGE IN LOG HOURLY WAGES			
	Weighted by Population		Excludes Zones with Highest Exposure	Unweighted	Weighted by Population		Excludes Zones with Highest Exposure	Unweighted
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	A. Past Change in Labor Market Outcomes, 1970–90							
Exposure to robots in subsequent period, 1993–2007	−.014	.006	−.158	−.217	.106	.145	.566	−.097
	(.077)	(.070)	(.174)	(.151)	(.242)	(.243)	(.453)	(.359)
Observations	722	722	712	722	59,230	59,230	58,402	59,230
R^2	.41	.43	.44	.32	.50	.50	.49	.30
	B. Estimates Controlling for Exposure to Industry Trends, 1970–90							
Exposure to robots	−.399	−.411	−.492	−.395	−.819	−.824	−.614	−.719
	(.046)	(.049)	(.138)	(.100)	(.136)	(.135)	(.273)	(.226)
Exposure to industry trends 1970–90	−.041	−.041	−.037	−.063	−.065	−.067	−.077	−.115
	(.019)	(.018)	(.019)	(.019)	(.034)	(.034)	(.034)	(.034)
Observations	722	722	712	722	87,100	87,100	85,776	87,100
R^2	.67	.67	.66	.63	.33	.33	.33	.08

(a) Table 4, part 1: Testing and controlling for pretrends.

	C. Estimates Controlling for the Change in the Dependent Variable, 1970–90							
Exposure to robots	−.434	−.449	−.587	−.551	−.904	−.913	−.789	−1.157
	(.057)	(.060)	(.132)	(.123)	(.144)	(.142)	(.291)	(.211)
Observations	722	722	712	722	55,988	55,988	55,182	55,988
R^2	.66	.67	.66	.62	.41	.41	.40	.16
	Covariates							
Census divisions	✓	✓	✓	✓	✓	✓	✓	✓
Demographics and industry shares	✓	✓	✓	✓	✓	✓	✓	✓
Trade, routine jobs		✓	✓	✓		✓	✓	✓

NOTE.—This table presents estimates of the effects of exposure to robots on past employment and wages, as well as estimates on contemporary changes in employment and wages controlling for past changes in these outcomes. Panel A presents estimates for changes in the employment-to-population ratio (cols. 1–4) and log hourly wages (cols. 5–8) between 1970 and 1990. For comparison with our main results, these outcomes are scaled to a 14-year equivalent change. Panel B presents long-differences estimates for changes in the employment-to-population ratio (cols. 1–4) and log hourly wages (cols. 5–8) between 1990 and 2007 controlling for a measure of exposure to industries that were in decline between 1970 and 1990 (see the main text for details on this variable). Panel C presents long-differences estimates for changes in the employment-to-population ratio (cols. 1–4) and log hourly wages (cols. 5–8) between 1990 and 2007 controlling for the change in the dependent variable between 1970 and 1990. The specifications for log hourly wages are estimated at the demographic cell × commuting zone level, where demographic cells are defined by age, gender, education, and race. Columns 1–3 and 5–7 present regressions weighted by population in 1990. Columns 3 and 7 present results excluding the top 1% of commuting zones with highest exposure to robots. Columns 4 and 8 present unweighted regressions. The covariates included in each model are reported at the bottom of the table. Columns 1 and 5 include census division dummies, demographic characteristics of commuting zones (log population; the share of females; the share of the population over 65 years old; the shares of the population with no college, some college, college or professional degree, and masters or doctoral degree; and the shares of whites, blacks, Hispanics, and Asians), the shares of employment in manufacturing and light manufacturing, and the female share of manufacturing employment. In panel A, the baseline covariates are from 1970, and in panels B and C they are from 1990. Columns 2–4 and 6–8 add exposure to Chinese imports and the share of employment in routine jobs. Standard errors that are robust against heteroskedasticity and correlation within states are given in parentheses.

(b) Table 4, part 2: Additional pretrend controls and notes.

6.8 Table 5 and Table 6: Automotive industry, capital deepening, IT, and value added

Read:

- Table 5 separates exposure to robots in the automotive industry from exposure to robots in other industries.
- Both sets of exposure have negative effects, and the tests generally do not reject equality of the coefficients.
- Table 6 controls for exposure to overall capital accumulation, IT capital, and industry value added.
- The robot coefficient remains negative and significant after these controls.

Economic message: The results are not only an automotive-industry story and are not just a proxy for capital deepening or IT investment. Robots appear to have a distinctive labor-displacing effect.

TABLE 5
EFFECTS OF ROBOTS ON EMPLOYMENT AND WAGES:
THE ROLE OF THE AUTOMOTIVE INDUSTRY

	Long Differences, 1990–2007			Stacked Differences, 1990–2000 and 2000–2007		
	Weighted by Population		Unweighted	Weighted by Population		Unweighted
	(1)	(2)	(3)	(4)	(5)	(6)
	A. Change in Employment-to-Population Ratio					
Exposure to robots in automotive industry	−.429	−.459	−.571	−.620	−.566	−.771
	(.078)	(.065)	(.153)	(.088)	(.054)	(.127)
Exposure to robots in other industries	−.505	−.370	−.451	−.654	−.449	−.695
	(.210)	(.117)	(.133)	(.212)	(.181)	(.161)
Test for equality of coefficients (p-value)	.69	.42	.47	.87	.54	.72
Observations	722	722	722	1,444	1,444	1,444
R^2	.27	.67	.62	.24	.41	.39
	B. Change in Log Hourly Wages					
Exposure to robots in automotive industry	−1.196	−.907	−.914	−1.507	−1.486	−1.740
	(.105)	(.118)	(.225)	(.162)	(.144)	(.282)
Exposure to robots in other industries	−1.314	−.715	−.955	−1.740	−1.144	−1.588
	(.515)	(.331)	(.377)	(.710)	(.741)	(.595)
Test for equality of coefficients (p-value)	.80	.52	.93	.73	.63	.81
Observations	87,100	87,100	87,100	183,606	183,606	183,606
R^2	.32	.33	.08	.28	.29	.09
	Covariates					
Time period dummies				✓	✓	✓
Census divisions	✓	✓	✓	✓	✓	✓
Baseline covariates		✓	✓		✓	✓

NOTE.—This table presents estimates of the effects of exposure to robots separately for the automotive industry and other industries. The p-value for a test of equality between the coefficients of exposure to robots in the automotive industry and in other industries is reported below the estimates. Columns 1–3 present long-differences estimates for the 1990–2007 period. Columns 4–6 present stacked-differences estimates for the 1990–2000 and 2000–2007 periods. Panel A presents results for the employment-to-population ratio. Panel B presents results for log hourly wages. The specifications in panel B are estimated at the demographic cell × commuting zone level, where demographic cells are defined by age, gender, education, and race. Columns 1, 2, 4, and 5 present regressions weighted by population in 1990. Columns 3 and 6 present unweighted regressions. The covariates in each model are reported at the bottom of the table. Columns 1 and 4 include only census division dummies (and time period dummies in the stacked-differences specifications). Columns 2, 3, 5, and 6 add demographic characteristics of commuting zones (log population; the share of females; the share of the population over 65 years old; the shares of the population with no college, some college, college or professional degree, and masters or doctoral degree; and the shares of whites, blacks, Hispanics, and Asians), the shares of employment in manufacturing

(a) Table 5: Role of the automotive industry.

TABLE 6
EFFECTS OF ROBOTS CONTROLLING FOR CAPITAL DEEPENING, IT CAPITAL, AND VALUE ADDED

	Change in Employment-to-Population Ratio				Change in Log Hourly Wages			
	Long Differences		Stacked Differences		Long Differences		Stacked Differences	
	Weighted (1)	Unweighted (2)	Weighted (3)	Unweighted (4)	Weighted (5)	Unweighted (6)	Weighted (7)	Unweighted (8)
	A. Estimates Controlling for Exposure to Capital							
Exposure to robots	−.435	−.495	−.502	−.601	−.822	−.827	−1.299	−1.407
	(.056)	(.114)	(.057)	(.083)	(.137)	(.213)	(.154)	(.258)
Exposure to capital	.025	.027	.065	.111	.116	.137	.200	.221
	(.020)	(.021)	(.036)	(.028)	(.033)	(.031)	(.099)	(.073)
Observations	722	722	1,444	1,444	87,100	87,100	183,606	183,606
R^2	.67	.62	.41	.41	.33	.08	.29	.09
	B. Estimates Controlling for Exposure to IT Capital							
Exposure to robots	−.445	−.515	−.507	−.734	−.854	−.921	−1.344	−1.627
	(.058)	(.116)	(.053)	(.092)	(.135)	(.199)	(.202)	(.298)
Exposure to IT capital	.006	.009	.054	.014	.067	.066	.123	.089
	(.009)	(.010)	(.032)	(.013)	(.017)	(.016)	(.050)	(.038)
Observations	722	722	1,444	1,444	87,100	87,100	183,606	183,606
R^2	.67	.62	.42	.39	.34	.08	.29	.09

(b) Table 6, part 1: Controls for capital and IT capital.

6.9 Table 6 continued and Table 7: IV estimates

Read:

- The continuation of Table 6 adds exposure to industry value added and again finds robust negative effects of robots.
- Table 7 reports IV estimates using US robot exposure instrumented by the European exposure measure.
- The first-stage coefficients are strong, and the IV estimates remain negative for both employment and wages.
- The IV results support the interpretation that the estimated effects are tied to robot technology rather than measurement error or endogenous US adoption.

Economic message: The IV results reinforce the reduced-form findings and help translate local estimates into aggregate labor market effects.

	C. Estimates Controlling for Exposure to Industry Value Added							
Exposure to robots	−.483	−.563	−.588	−.782	−.903	−.958	−1.487	−1.717
	(.061)	(.113)	(.051)	(.087)	(.133)	(.201)	(.183)	(.292)
Exposure to industry value added	.060	.064	.082	.079	.034	.033	.097	.062
	(.013)	(.015)	(.024)	(.018)	(.021)	(.017)	(.054)	(.025)
Observations	722	722	1,444	1,444	87,100	87,100	183,606	183,606
R^2	.69	.64	.42	.41	.33	.08	.29	.09
	Covariates							
Time period dummies			✓	✓			✓	✓
Census divisions and baseline covariates	✓	✓	✓	✓	✓	✓	✓	✓

Note.—This table presents estimates of the effects of exposure to robots, capital accumulation, IT capital, and value added on employment and wages Columns 1, 2, 5, and 6 present long-differences estimates for 1990–2007. Columns 3, 4, 7, and 8 present stacked-differences estimates for 1990–2000 anc 2000–2007. Columns 1–4 present results for the employment-to-population ratio. Columns 5–8 present results for log hourly wages. The specifications ir cols. 5–8 for log hourly wages are estimated at the demographic cell × commuting zone level, where demographic cells are defined by age, gender, ed ucation, and race. Odd-numbered columns present regressions weighted by population in 1990. Even-numbered columns present unweighted regres sions. In panel A, we control for a measure of exposure to capital, constructed by interacting the increase in log capital in each industry with its baseline employment share in the commuting zone. In panel B, we control for a measure of exposure to IT capital, constructed by interacting the increase in log IT capital in each industry with its baseline employment share in the commuting zone. In panel C, we control for a measure of exposure to industry value added, constructed by interacting the increase in log value added in each industry (between 1992 and 2007 in all models) with its baseline employmen share in the commuting zone. All models include census division dummies (and time period dummies in the stacked-differences specifications), demo graphic characteristics of commuting zones (log population; the share of females; the share of the population over 65 years old; the shares of the pop ulation with no college, some college, college or professional degree, and masters or doctoral degree; and the shares of whites, blacks, Hispanics, anc Asians), the shares of employment in manufacturing and light manufacturing, the female share of manufacturing employment, exposure to Chinese im ports, and the share of employment in routine jobs. Standard errors that are robust against heteroskedasticity and correlation within states are given ir parentheses.

(a) Table 6, part 2: Controls for industry value added.

Effects of Robots on Employment and Wages: IV Estimates

	Change in Employment-to-Population Ratio			Change in Log Hourly Wages		
	(1)	(2)	(3)	(4)	(5)	(6)
	A. Long Differences, 1990–2007					
US exposure to robots	−.375	−.377	−.388	−1.022	−.762	−.768
	(.119)	(.088)	(.091)	(.225)	(.147)	(.148)
Observations	722	722	722	87,100	87,100	87,100
First-stage coefficient	1.19	1.15	1.15	1.19	1.15	1.15
First-stage F-statistic	38.16	32.19	33.62	39.81	33.17	34.35
	B. Alternative Imputation of US Data, 1990–2007					
US exposure to robots	−.388	−.391	−.402	−1.060	−.790	−.796
	(.123)	(.092)	(.094)	(.234)	(.153)	(.153)
Observations	722	722	722	87,100	87,100	87,100
First-stage coefficient	1.15	1.11	1.11	1.15	1.11	1.11
First-stage F-statistic	38.16	32.19	33.62	39.81	33.17	34.35
	C. Long Differences, 1990–2014					
US exposure to robots	−.303	−.241	−.250	−1.268	−1.103	−1.128
	(.126)	(.077)	(.080)	(.169)	(.165)	(.170)
Observations	722	722	722	90,341	90,341	90,341
First-stage coefficient	1.20	1.14	1.15	1.21	1.14	1.15
First-stage F-statistic	105.01	68.75	73.34	110.32	70.35	74.86
	D. Long Differences, 2000–2007					
US exposure to robots	−.623	−.574	−.585	−1.376	−1.147	−1.191
	(.137)	(.067)	(.067)	(.227)	(.172)	(.176)
Observations	722	722	722	99,319	99,319	99,319
First-stage coefficient	.79	.76	.75	.79	.76	.75
First-stage F-statistic	211.46	123.73	124.32	223.26	131.40	131.83
	E. Long Differences, 2000–2014					
US exposure to robots	−.451	−.327	−.339	−1.590	−1.566	−1.601
	(.190)	(.065)	(.067)	(.156)	(.183)	(.191)
Observations	722	722	722	106,375	106,375	106,375
First-stage coefficient	1.00	.92	.93	1.00	.92	.93
First-stage F-statistic	1,195.67	305.25	298.98	1,296.85	324.38	316.40
	Covariates					
Division dummies	✓	✓	✓	✓	✓	✓
Demographics and industry shares		✓	✓		✓	✓
Trade, routine jobs			✓			✓

Note.—This table presents IV estimates of the effects of exposure to robots on employment and wages for different time periods. Panels A and B present results for 1990–2007. Panel C presents results for 1990–2014. Panel D presents results for 2000–2007. Panel E presents results for 2000–2014. In all models, we instrument the US exposure to robots using exposure to robots from *EURO5*. In panels A and C–E, we rescale the US exposure to robots to match the time period used. In panel B, we use an alternative imputation strategy for US exposure to robots described in the main text. Columns 1–3 present results for the employment-to-population ratio. Columns 4–6 present results for log hourly wages. The specifications for log hourly wages are estimated at the demographic cell × commuting zone level, where demographic cells are defined by age, gender, education, and race. All IV estimates are from regressions weighted by population in 1990. The covariates included in each model are reported at the bottom of the table. Columns 1 and 4 include only census division dummies. Columns 2 and 5 add demographic characteristics of commuting zones (log population; the share of females; the share of the population over 65 years old; the

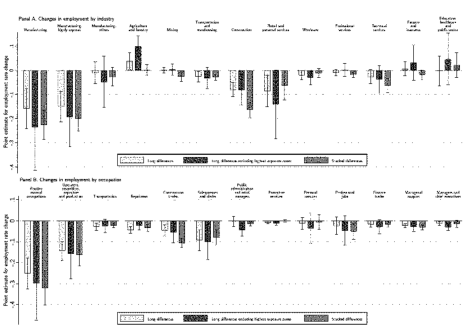
(b) Table 7: IV estimates.

6.10 Figure 8 and Figure 9: Heterogeneous effects by industry, occupation, education, and gender

Read:

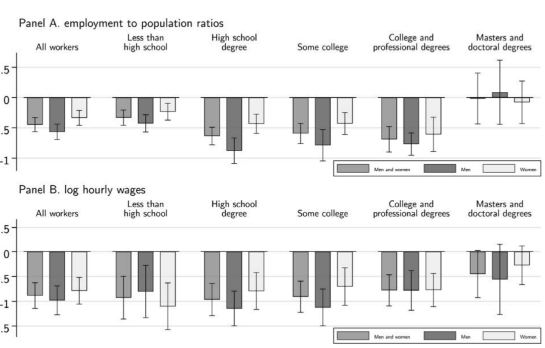
- Figure 8 shows that employment losses are concentrated in manufacturing and in occupations that are routine, manual, blue-collar, assembly, and production-related.
- Nontradable sectors such as construction and retail also experience negative effects, consistent with local demand spillovers.
- Figure 9 shows negative employment and wage effects for broad worker groups, with particularly visible effects for men and for groups without advanced education.

Economic message: Robot exposure has distributional consequences. It especially harms workers whose tasks are close substitutes for industrial robots and affects local service sectors through reduced local demand.



;. 8.—Effects of robots on industries and occupations. This figure presents estimates of the effects of exposure to robots on changes in industry oyment-to-population ratios (*A*) and changes in occupation employment-to-population ratios (*B*). The capped lines provide 95% confidence inter- The first set of estimates are from long-differences specifications as in column 4 of table 2. The second set of estimates are from long-differences fications as in column 5 of table 2 (where we remove the top 1% of commuting zones with the highest exposure to robots). The third set of estimates om stacked-differences specifications as in column 4 of table 3.

(a) Figure 8: Effects by industry and occupation.



9.—Effects of robots on employment and wages by education and gender. This figure presents estimates of the effects of exposure to robots on changes in ployment-to-population ratio (*A*) and changes in log hourly wages (*B*) for all workers and for men and women with different education levels (less than high high school degree, some college, college or professional degree, and masters or doctoral degree). The capped lines provide 95% confidence intervals.

(b) Figure 9: Effects by education and gender.

6.11 Figure 10: Wage distribution

Read:

- Figure 10 estimates effects of robot exposure across wage quantiles.
- The effects are generally negative across the distribution, rather than being limited to a single narrow wage group.
- The panels separate all workers, workers without college, workers with college, men, and women.

Economic message: Robots affect the wage distribution broadly, but the detailed pattern differs across education and gender groups. This reinforces that robots are not just a sector-level employment shock; they also reshape the distribution of labor market outcomes.

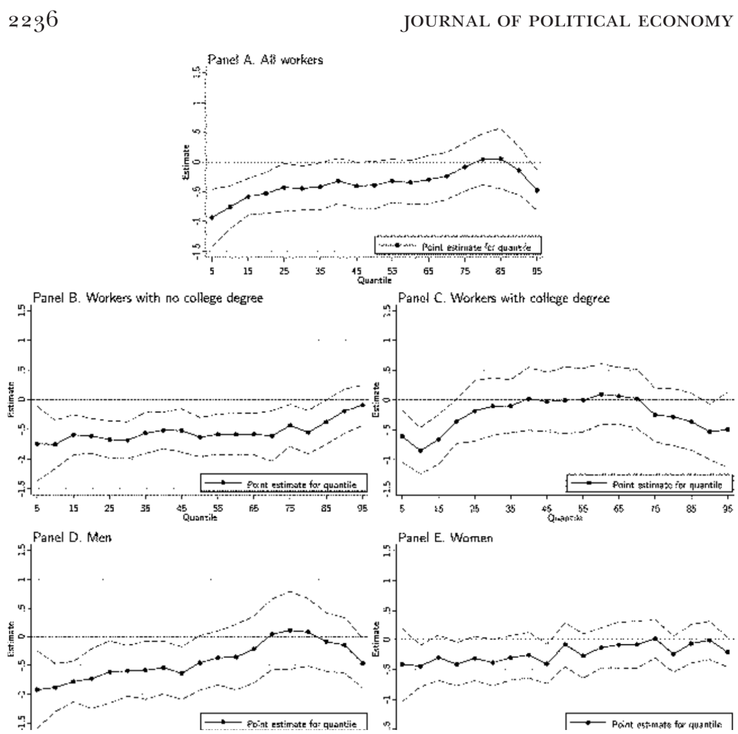


Figure 10: Effects of robots on the wage distribution.

7 Short summary

- The paper studies how industrial robots affect US local labor markets.
- It builds a task-based model where robots displace workers in some tasks but also raise productivity.
- Exposure to robots is measured by combining baseline local industry shares with industry-level robot penetration.
- The preferred source of robot technology variation is European robot adoption in countries ahead of the United States.

- Robot-exposed commuting zones experience lower employment-to-population ratios and lower wages.
- The estimates are robust to controls for demographics, manufacturing shares, Chinese imports, routine jobs, pretrends, automotive exposure, capital deepening, IT capital, and value added.
- Negative effects are concentrated in manufacturing and routine/manual occupations, but local spillovers also affect nontradable sectors.
- The paper interprets robots as a distinctive automation technology, not merely as another form of capital accumulation.

8 Conclusion

8.1 Core conclusion

Acemoglu and Restrepo show that industrial robots have economically meaningful negative effects on US employment and wages. Areas more exposed to robot-adopting industries experience worse labor market outcomes after 1990, and these results are not explained by preexisting trends, trade shocks, the automotive sector alone, or general capital deepening.

8.2 Economic intuition

The key intuition is the displacement effect. When robots take over tasks previously performed by workers, labor demand falls in those tasks. Although robot adoption also lowers costs and raises productivity, the paper's estimates imply that the displacement effect dominates for US local labor markets during the sample period.

8.3 Incremental contribution revisited

The paper advances the automation literature in three main ways. First, it uses direct data on industrial robots rather than forecasts of automation risk. Second, it links a task-based theory to a concrete shift-share empirical design. Third, it quantifies both local and aggregate effects, showing that robot exposure has sizable local costs even after accounting for broader equilibrium spillovers.

8.4 Implications for labor markets and policy

The findings imply that automation can generate concentrated local adjustment costs. Policies that treat technological change as uniformly beneficial may miss these distributional effects. The most exposed workers are those in robot-substitutable tasks, especially manufacturing and routine manual occupations. Adjustment policies may therefore need to focus on retraining, mobility, and local labor market support.

8.5 Limitations

The analysis focuses on industrial robots, not all forms of automation or artificial intelligence. The shift-share design depends on the validity of the industry-level robot adoption shocks. The estimates identify medium-run effects over the sample period and may not fully capture longer-run creation of new tasks, changes in product design, firm entry, or endogenous innovation responses.

8.6 Takeaway

The central takeaway is that robots are a labor-displacing technology with measurable consequences for US workers. They raise productivity in robot-adopting industries, but they also reduce employment and wages in exposed local labor markets. The paper makes the debate over automation concrete by showing where, how, and for whom robot adoption matters.